

Course name and code: Web and Mobile Development (IT645)

Course placement: core course for M.Sc.(IT) Sem III

Course credits: L-T-P (3-0-2-4)

Course instructor: Shruti Bhilare

Course objective: The objective of this course is to provide an overview of the building blocks of a functional webpage and to understand the frontend and backend technologies that power the modern websites and mobile applications. It will help learn skills to develop dynamic webpages and integrate APIs through Javascript-enabled frontend framework, ReactJS and backend of NodeJS. It aims at teaching a hybrid mobile application development technology, React Native for iOS as well as Android operating systems.

Course content: The basic building blocks of webpages: HTML5, CSS3 and Javascript. Additionally, front-end framework for creating dynamic web pages (ReactJS) and the NodeJS and MongoDB backend comprising the MERN stack and React Native. Detailed course content is listed below.

Prerequisites: HTML, CSS, programming skills,

Lecture schedule

S. no.	Description		No. of lectures
1	Overview of HTML5		3
	1.1	Selected topics in HTML	
	1.2	HTML: document structure, metadata and scripting, sectioning and layout, text content, forms and input, images and multimedia, tables	
2	Overview of CSS3		5
	2.1	Selected topics in CSS	
	2.2	Text and Font styles, Box model, Backgrounds, Flexbox, Grid, positioning, lists, tables, effects, transitions and animations	
3	Javascript (Native)		6
	3.1	Variables, operators, functions: arrow functions,	
	3.2	Higher order functions,	
	3.3	Arrays and objects, OOP,	
	3.4	Asynchronous programming: callbacks, promises, async await,	
	3.5	Fetch API, Geolocation API	
	3.6	Event loop	
	3.7	Event handling	
4	Front end framework (ReactJS)		6
	4.1	Introduction to ReactJS, React components, props and render(), lifecycle and lifecycle methods,	
	4.2	Build a simple React application	

	4.3	Rendering and updating elements	
	4.4	Javascript XML (JSX)	
	4.5	Component intercommunication	
	4.6	Event handling	
		Backend	
5		Server side scripting using NodeJS	6
	5.1	Server side programming basics, NodeJS introduction, Node commands	
	5.2	Node runtime environment, Installation: package.json and dependencies, Node modules: HTTP, BodyParser, nodemon etc	
	5.3	Fetch API, HTTP GET and POST requests, query parameters, route parameters, JSON	
6		MongoDB	3
	6.1	Introduction to document-oriented database, mongo shell commands,	
	6.2	using MongoDB in a server, using callbacks and promises	
	6.3	DB operations: search, insert, update, delete.	
7		Full stack web development (MERN)	2
	7.1	Putting it all together	
8		React Native	8
	8.1	Introduction to React Native, Setup	
	8.2	workflow, UI and interaction, connectivity, platform APIs	
	8.3	Modules, Deployment	

Suggested textbooks and references

- 1 Richard Clark, "Beginning HTML5 and CSS3", Apress Publishers.
2. Marijn Haverbeke, "Eloquent JavaScript: A Modern Introduction to Programming", MIT press.
3. Jason Krol, Bruno Joseph D'mello, Mithun Satheesh, "Web Development with MongoDB and NodeJS-second edition", O'Reilly.
4. Bonnie Eisenman, "Learning React Native: Building Mobile Applications with Javascript", O'Reilly
4. <https://www.mongodb.com/>
5. <https://nodejs.org/en/>
6. <https://reactjs.org/>
7. <https://expressjs.com/>
8. <https://reactnative.dev/docs/getting-started>

Mapping to program outcomes of NBA

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X	X		X				X			

Grading Policy

- Insem 1: 15%
- Insem 2: 20%
- Endsem exam: 25%
- Lab Assignments: 20%
- Project work: 20%

Course Outcomes

- Generation of awareness of current web and mobile technology landscape for creating full stack applications
- Implementation of feature-rich applications from scratch
- In-depth understanding of MERN stack to create a functional web application
- In-depth understanding of React Native to create hybrid mobile application